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COMPREHENSIVE STATISTICAL METHODS SUMMARY
When to Use Which Test, Assumptions, and Formulas
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1. Introduction to Empirical Statistics
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1. INTRODUCTION TO EMPIRICAL STATISTICS
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SCALES OF MEASUREMENT
--------------------------------------------------------------------------------
Scale       Properties                      Examples              Valid Operations
--------------------------------------------------------------------------------
Nominal     Categories, no order           Gender, color         Frequency, mode, χ²
Ordinal     Categories with order          Likert scales         Median, percentiles
Interval    Equal intervals, no true zero  Temperature (°C)      Mean, SD, correlation
Ratio       Equal intervals, true zero     Height, weight, time  All operations
Continuous  Infinite possible values       Reaction time         Mean, SD, regression
Discrete    Countable values               Number of errors      Counts, proportions
--------------------------------------------------------------------------------

KEY CONCEPTS:
• Reliability: Consistency of measurement (test-retest, internal consistency)
• Validity: Measuring what you intend to measure
  - Content validity: Covers the construct
  - Criterion validity: Correlates with gold standard
  - Construct validity: Fits theoretical framework

• Post-hoc testing: Testing after seeing data → inflates Type I error
• Data mining: Finding patterns by chance in large datasets
• Publication bias: Only significant results get published
• Confounds: Variables that provide alternative explanations

RULE: Always correct for multiple comparisons when doing post-hoc tests!
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2. DATA EXPLORATION AND DISTRIBUTIONS
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MEASURES OF CENTRAL TENDENCY
--------------------------------------------------------------------------------
Measure   Formula                     When to Use              Influenced by Outliers?
--------------------------------------------------------------------------------
Mean      μ = Σx / n                 Symmetric distributions   YES - very sensitive
Median    Middle value when sorted   Skewed distributions      NO - robust
Mode      Most frequent value        Categorical data          NO - robust
--------------------------------------------------------------------------------

MEASURES OF VARIABILITY
--------------------------------------------------------------------------------
Measure         Formula                          Interpretation
--------------------------------------------------------------------------------
Range           max - min                        Spread of data
IQR             Q3 - Q1                          Middle 50% spread
Variance        σ² = Σ(x - μ)² / n              Average squared deviation
                s² = Σ(x - x̄)² / (n-1)          (sample uses n-1)
Std Deviation   σ = √(σ²)                        Average deviation (same units)
Std Error       SE = σ / √n                      Uncertainty of mean estimate
--------------------------------------------------------------------------------

NORMAL DISTRIBUTION
--------------------------------------------------------------------------------
Properties:
• Bell-shaped, symmetric around mean
• Mean = Median = Mode
• 68% of data within ±1 SD
• 95% of data within ±1.96 SD
• 99.7% of data within ±3 SD

Z-score (Standardization):
z = (x - μ) / σ

Interpretation: How many standard deviations from the mean

Probability from Z-score:
P(X ≤ x) = Φ(z) → Look up in standard normal table

CENTRAL LIMIT THEOREM
--------------------------------------------------------------------------------
Statement: The sampling distribution of the mean approaches a normal
distribution as sample size increases, regardless of the population distribution.

Formula: X̄ ~ N(μ, σ²/n)
```

Where: μ = population mean, σ^2/n = variance of sampling distribution

Practical rule: $n \geq 30$ usually sufficient for CLT to apply

Implications:
- Can use parametric tests even if population not normal (large n)
- Standard error decreases with $\sqrt{n}$
- Can construct confidence intervals for means

VISUALIZATION TYPES

Plot Type	Use Case	What It Shows
Histogram	Distribution of continuous	Frequency distribution
Density plot	Smooth distribution	Probability density
Bar plot	Categorical frequencies	Count or proportion per category
Boxplot	Distribution + outliers	Median, IQR, outliers
Q-Q plot	Check normality	Observed vs expected quantiles
Scatterplot	Relationship between 2 vars	Correlation, patterns

How to read Q-Q plot:
- Points on diagonal line → Normal distribution
- Points curve above line → Right skew (heavy right tail)
- Points curve below line → Left skew (heavy left tail)
- S-shaped pattern → Heavy tails (platykurtic)

3. HYPOTHESIS TESTING FUNDAMENTALS

HYPOTHESIS TESTING FRAMEWORK

H_0 (Null Hypothesis): No effect, no difference (status quo)
H_1 (Alternative): There is an effect/difference

Decision Rules:
- If p-value $< \alpha$: Reject H_0 (statistically significant)
- If p-value $\geq \alpha$: Fail to reject H_0 (not significant)

α (alpha level): Typically 0.05 (5% significance level)

TYPES OF ERRORS

	H_0 is TRUE	H_0 is FALSE
	(Reality: No effect)	(Reality: Effect exists)

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--------------------------------------------------------------------------------
Reject H₀           Type I Error (α)      Correct Decision (Power)
(Say effect exists) False Positive        True Positive

Fail to Reject H₀   Correct Decision      Type II Error (β)
(Say no effect)     True Negative         False Negative (Miss)
--------------------------------------------------------------------------------

TYPE I ERROR (α): Probability of rejecting H₀ when it's true
• Controlled by setting α level (usually 0.05)
• "False positive" - claiming effect when none exists

TYPE II ERROR (β): Probability of failing to reject H₀ when it's false
• Depends on: sample size, effect size, α level, variability
• "False negative" - missing a real effect

STATISTICAL POWER = 1 - β
• Probability of correctly rejecting H₀ when it's false
• Probability of detecting a real effect
• Typically aim for power ≥ 0.80 (80%)

Factors increasing power:
1. Larger sample size (n↑)
2. Larger effect size (stronger effect)
3. Less variability in data (σ↓)
4. Higher α level (more liberal)
5. One-tailed test (if direction predicted)

ONE-TAILED VS TWO-TAILED TESTS
--------------------------------------------------------------------------------
Two-tailed (non-directional):  H₁: μ ≠ μ₀
• Tests for difference in EITHER direction
• More conservative (harder to reach significance)
• Use when: No prediction about direction

One-tailed (directional):      H₁: μ > μ₀  or  H₁: μ < μ₀
• Tests for difference in ONE specific direction
• More powerful (easier to reach significance)
• Use when: Strong theoretical prediction about direction
• Risk: Miss effects in opposite direction

Critical value relationship:
• Two-tailed α=.05 → critical value more extreme than one-tailed α=.05
• One-tailed α=.05 = Two-tailed α=.10

CONFIDENCE INTERVALS
--------------------------------------------------------------------------------
Formula (for mean): CI = x̄ ± (t_critical × SE)
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                      = x̄ ± (t_critical × s/√n)

Where:
• x̄ = sample mean
• t_critical = critical value from t-distribution
• s = sample standard deviation
• n = sample size

Common confidence levels:
• 90% CI: t_critical ≈ 1.645 (for large n)
• 95% CI: t_critical ≈ 1.96 (for large n)
• 99% CI: t_critical ≈ 2.576 (for large n)

Interpretation (FREQUENTIST):
"If we repeated this study 100 times, 95 of the intervals would contain
the true population mean."

NOT: "There's a 95% probability the true mean is in this interval"

Relationship to hypothesis testing:
• If 95% CI excludes null value (e.g., 0) → significant at α=.05
• If 95% CI includes null value → not significant

PARAMETRIC ASSUMPTIONS
--------------------------------------------------------------------------------
Most parametric tests (t-test, ANOVA, regression) assume:

1. NORMALITY: Data (or residuals) are normally distributed
   Check: Shapiro-Wilk test, Q-Q plots, histograms
   Robust to violations if: n ≥ 30 (CLT)

2. INDEPENDENCE: Observations are independent of each other
   Check: Consider study design
   Violation: Use repeated measures / mixed models

3. HOMOGENEITY OF VARIANCE (Homoscedasticity): Equal variances across groups
   Check: Levene's test, Residual plots
   Solution: Use Welch's t-test, robust methods

4. CONTINUOUS DV: Dependent variable is continuous (interval/ratio)
   Violation: Use non-parametric tests or GLM

5. NO OUTLIERS: Or outliers don't unduly influence results
   Check: Boxplots, Cook's distance
   Solution: Remove outliers, use robust methods, transform data

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4. TESTS FOR CATEGORICAL DATA
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DECISION TREE FOR CATEGORICAL DATA
--------------------------------------------------------------------------------
Question: What type of categorical data?

→ One categorical variable, test against known proportion
  → Use: BINOMIAL TEST

→ Two categorical variables, test for association/independence
  → Are samples independent?
     → YES: Use CHI-SQUARE TEST or FISHER'S EXACT TEST
     → NO (paired/repeated): Use McNEMAR'S TEST
--------------------------------------------------------------------------------

4.1 BINOMIAL TEST
--------------------------------------------------------------------------------
USE WHEN:
• One categorical variable with 2 outcomes (success/failure)
• Testing if proportion differs from a specific value
• Small sample size (exact test)

SCENARIO EXAMPLES:
• Is a coin fair? (H₀: p = 0.5)
• Do patients prefer treatment A over B? (H₀: p = 0.5)
• Is success rate better than chance (25%)? (H₀: p = 0.25)

ASSUMPTIONS:
✓ Binary outcomes
✓ Independent trials
✓ Constant probability across trials

FORMULAS:
Probability of exactly k successes in n trials:
P(X = k) = C(n,k) × p^k × (1-p)^(n-k)

Where: C(n,k) = n! / (k!(n-k)!)

Test statistic (large n, normal approximation):
z = (p̂ - p₀) / √(p₀(1-p₀)/n)

Where:
• p̂ = observed proportion = k/n
• p₀ = null hypothesis proportion
• n = sample size

R CODE:
binom.test(x = successes, n = total, p = null_prob)
```

INTERPRETATION:
p-value < α → Reject H_0, proportion differs from null value

4.2 CHI-SQUARE TEST OF INDEPENDENCE (χ^2)

USE WHEN:
• Two categorical variables
• Testing for association/independence between variables
• Data in contingency table (crosstab)
• Expected frequencies ≥ 5 in each cell

SCENARIO EXAMPLES:
• Is smoking status related to gender?
• Does treatment type affect recovery (yes/no)?
• Is movie preference associated with age group?

ASSUMPTIONS:
✓ Categorical variables
✓ Independent observations (each person in only one cell)
✓ Expected frequency ≥ 5 in at least 80% of cells
✓ Expected frequency ≥ 1 in all cells
✓ Sufficiently large sample size

FORMULAS:
Chi-square test statistic:
$\chi^2 = \Sigma [(O - E)^2 / E]$

Where:
• O = Observed frequency in each cell
• E = Expected frequency in each cell

Expected frequency:
E_ij = (Row_i total × Column_j total) / Grand total

Degrees of freedom:
df = (r - 1) × (c - 1)

Where: r = number of rows, c = number of columns

Effect size (Cramér's V):
$V = \sqrt{\chi^2 / (n \times \min(r-1, c-1))}$

Interpretation: 0 = no association, 1 = perfect association
Small: 0.1, Medium: 0.3, Large: 0.5

```
R CODE:
chisq.test(table(var1, var2))

DECISION:
p-value < α → Reject H₀, variables are associated (not independent)

POST-HOC TESTS:
If χ² is significant, examine:
• Standardized residuals: (O - E) / √E
• Values > |2| indicate cells contributing to association

--------------------------------------------------------------------------------

4.3 FISHER'S EXACT TEST
--------------------------------------------------------------------------------
USE WHEN:
• Two categorical variables (like χ²)
• BUT: Small sample size
• Expected frequencies < 5 in any cell
• Provides exact p-value (not approximation)

SCENARIO EXAMPLES:
• Same as χ², but with small samples
• Clinical trials with few patients
• Rare events

ASSUMPTIONS:
✓ Categorical variables
✓ Independent observations
× NO minimum sample size requirement (this is the advantage!)

FORMULA:
For 2×2 table:
       Success  Failure  Total
Group1    a        b      a+b
Group2    c        d      c+d
Total    a+c      b+d      n

p = [C(a+b, a) × C(c+d, c)] / C(n, a+c)

Where C(n,k) = n! / (k!(n-k)!)

R CODE:
fisher.test(table(var1, var2))

WHEN TO USE FISHER vs CHI-SQUARE:
• Expected frequency < 5 in >20% of cells → Use Fisher's Exact
• All expected frequencies ≥ 5 → Use Chi-square (more efficient)
```

• Very large sample (n > 1000) → Chi-square (Fisher's can be slow)

4.4 McNEMAR'S TEST

USE WHEN:
• Two categorical variables (binary)
• PAIRED/REPEATED MEASURES (same subjects measured twice)
• Testing for change in proportions
• 2×2 contingency table with dependent samples

SCENARIO EXAMPLES:
• Before/after treatment: Did symptoms improve?
• Paired preferences: Do twins agree on choice?
• Test-retest: Did diagnosis change?

DESIGN:

	After: Yes	After: No
Before: Yes	a	b
Before: No	c	d

Focus on discordant pairs: b and c (changed responses)

ASSUMPTIONS:
✓ Binary categorical variables
✓ Paired observations (same subjects twice)
✓ Large enough sample of discordant pairs (b + c ≥ 10)

FORMULAS:
Test statistic (with continuity correction):
$\chi^2 = (|b - c| - 1)^2 / (b + c)$

Degrees of freedom: df = 1

Without continuity correction:
$\chi^2 = (b - c)^2 / (b + c)$

R CODE:
```
mcnemar.test(matrix(c(a, b, c, d), nrow=2))
```

INTERPRETATION:
Tests if proportions changed significantly
p-value < α → Significant change from before to after

Effect size (Odds Ratio for change):
OR = b / c

```
If OR > 1: More likely to change from No→Yes than Yes→No
If OR < 1: More likely to change from Yes→No than No→Yes

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SUMMARY TABLE: CATEGORICAL DATA TESTS
--------------------------------------------------------------------------------
Test            Variables       Design          Min Expected   Output
--------------------------------------------------------------------------------
Binomial        1 categorical   Single sample   None (exact)   Proportion test
Chi-square      2 categorical   Independent     ≥5 in cells    Association
Fisher's Exact  2 categorical   Independent     None (exact)   Association
McNemar's       2 categorical   Paired/Repeated ≥10 discordant Change test
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5. TESTS FOR COMPARING MEANS
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DECISION TREE FOR COMPARING MEANS
--------------------------------------------------------------------------------
Question: How many groups are you comparing?

→ ONE group (compare to known value)
  → Population SD known?
     YES → ONE-SAMPLE Z-TEST
     NO → ONE-SAMPLE T-TEST

→ TWO groups
  → Are they independent or paired?
     INDEPENDENT:
        → Equal variances?
           YES → STUDENT'S T-TEST
           NO → WELCH'S T-TEST
        → Normality violated? → MANN-WHITNEY U TEST

     PAIRED/REPEATED:
        → Normally distributed?
           YES → PAIRED T-TEST
           NO → WILCOXON SIGNED-RANK TEST

→ THREE OR MORE groups → See ANOVA section
--------------------------------------------------------------------------------

5.1 ONE-SAMPLE Z-TEST
--------------------------------------------------------------------------------
USE WHEN:
• Comparing sample mean to known population mean
```

• Population standard deviation (σ) is KNOWN
• Large sample size (n ≥ 30) OR population is normal

SCENARIO EXAMPLES:
• Is average IQ of sample different from population (μ=100, σ=15)?
• Does average height differ from national average (known σ)?

ASSUMPTIONS:
✓ Continuous dependent variable
✓ Population SD known
✓ Normal distribution OR n ≥ 30
✓ Independent observations

FORMULAS:
Test statistic:
$z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$

Where:
• $\bar{x}$ = sample mean
• μ_0 = population mean (null hypothesis value)
• σ = population standard deviation
• n = sample size

Critical values (two-tailed, α=.05): ±1.96
Critical values (one-tailed, α=.05): ±1.645

Confidence Interval:
CI = x̄ ± z_critical × (σ / √n)

R CODE:
```
# No built-in function, calculate manually:
z <- (mean(x) - mu0) / (sigma / sqrt(n))
p_value <- 2 * pnorm(-abs(z))  # two-tailed
```

DECISION:
$|z| > 1.96 \rightarrow$ Reject H_0 at α=.05 (two-tailed)

5.2 ONE-SAMPLE T-TEST

USE WHEN:
• Comparing sample mean to known value
• Population SD is UNKNOWN (estimate from sample)
• Small to medium sample size OR normal distribution

SCENARIO EXAMPLES:
• Is average reaction time different from 500ms?

• Does class average differ from passing grade (70)?
• Is mean score different from chance level?

ASSUMPTIONS:
✓ Continuous dependent variable
✓ Normal distribution (especially for small n)
✓ Independent observations
× Robust to normality violations if n ≥ 30

FORMULAS:
Test statistic:
$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$

Where:
• $\bar{x}$ = sample mean
• μ_0 = null hypothesis mean
• s = sample standard deviation
• n = sample size

Degrees of freedom: df = n - 1

Confidence Interval:
$CI = \bar{x} \pm t_critical \times (s / \sqrt{n})$

R CODE:
```
t.test(x, mu = mu0)                    # two-tailed
t.test(x, mu = mu0, alternative="greater")  # one-tailed
```

OUTPUT INTERPRETATION:
t(df) = t-value, p = p-value
95% CI: [lower, upper]

EFFECT SIZE (Cohen's d):
$d = (\bar{x} - \mu_0) / s$

Interpretation: Small: 0.2, Medium: 0.5, Large: 0.8

5.3 INDEPENDENT SAMPLES T-TEST (STUDENT'S)

USE WHEN:
• Comparing means of TWO INDEPENDENT groups
• Equal variances in both groups (homoscedasticity)
• Different participants in each group

SCENARIO EXAMPLES:
• Do males and females differ in height?

• Treatment group vs control group
• Morning people vs evening people on alertness

ASSUMPTIONS:
✓ Continuous dependent variable
✓ Two independent groups
✓ Normal distribution in both groups (or $n \geq 30$ per group)
✓ Equal variances (homoscedasticity)
✓ Independent observations

CHECK ASSUMPTIONS:
• Normality: Shapiro-Wilk test, Q-Q plots
• Equal variances: Levene's test, F-test ($F = s_1^2/s_2^2$)

FORMULAS:
Test statistic:
$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s^2_pooled \times (1/n_1 + 1/n_2))}$

Pooled variance:
$s^2_pooled = [(n_1-1)s_1^2 + (n_2-1)s_2^2] / (n_1 + n_2 - 2)$

Degrees of freedom: $df = n_1 + n_2 - 2$

Confidence Interval for difference:
$CI = (\bar{x}_1 - \bar{x}_2) \pm t_critical \times SE$

Where: $SE = \sqrt{(s^2_pooled \times (1/n_1 + 1/n_2))}$

R CODE:
```
t.test(DV ~ Group, data=data, var.equal=TRUE)
```

EFFECT SIZE (Cohen's d):
$d = (\bar{x}_1 - \bar{x}_2) / s_pooled$

Where: $s_pooled = \sqrt{(s^2_pooled)}$

Interpretation: Small: 0.2, Medium: 0.5, Large: 0.8

5.4 WELCH'S T-TEST (UNEQUAL VARIANCES)

USE WHEN:
• Comparing means of TWO INDEPENDENT groups
• Variances are UNEQUAL (heteroscedasticity)
• Levene's test significant → variances differ

SCENARIO EXAMPLES:

• Same as Student's t-test, but when equal variance assumption violated
• Often more robust and recommended as default

ASSUMPTIONS:
✓ Continuous dependent variable
✓ Two independent groups
✓ Normal distribution (or n ≥ 30)
× Does NOT assume equal variances

FORMULAS:
Test statistic:
$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$

Welch-Satterthwaite degrees of freedom (approximation):
$df = (s_1^2/n_1 + s_2^2/n_2)^2 / [(s_1^2/n_1)^2/(n_1-1) + (s_2^2/n_2)^2/(n_2-1)]$

R CODE:
```
t.test(DV ~ Group, data=data, var.equal=FALSE)  # Default in R
```

NOTE: This is the DEFAULT t.test() in R (var.equal=FALSE)

EFFECT SIZE:
Use same Cohen's d formula, but with pooled SD adjusted for unequal n:

$d = (\bar{x}_1 - \bar{x}_2) / \sqrt{[(s_1^2 + s_2^2) / 2]}$

5.5 PAIRED-SAMPLES T-TEST

USE WHEN:
• Comparing means of TWO RELATED groups
• Same participants measured twice (repeated measures)
• Matched pairs (e.g., twins, siblings)
• Pre-test vs post-test design

SCENARIO EXAMPLES:
• Before vs after treatment
• Left hand vs right hand performance
• First attempt vs second attempt
• Husband vs wife scores (matched pairs)

ASSUMPTIONS:
✓ Continuous dependent variable
✓ Paired observations (each score in group 1 matched to score in group 2)
✓ Differences are normally distributed
✓ Independent pairs (one pair's difference independent of another's)

FORMULAS:
First, calculate difference scores: $d_i = x_{1i} - x_{2i}$

Test statistic:
$t = (\bar{d} - 0) / (s_d / \sqrt{n})$

Where:
- $\bar{d}$ = mean of differences
- s_d = standard deviation of differences
- n = number of pairs

Degrees of freedom: $df = n - 1$

Confidence Interval:
$CI = \bar{d} \pm t_{critical} \times (s_d / \sqrt{n})$

R CODE:
```
t.test(x1, x2, paired=TRUE)
# or
t.test(Difference ~ 1, data=data)
```

EFFECT SIZE (Cohen's d for paired samples):
$d = \bar{d} / s_d$

Interpretation: Small: 0.2, Medium: 0.5, Large: 0.8

POWER ADVANTAGE:
Paired designs have more power because they control for individual differences
→ Reduces error variance

5.6 MANN-WHITNEY U TEST (WILCOXON RANK-SUM)

USE WHEN:
- Comparing TWO INDEPENDENT groups
- Non-parametric alternative to independent t-test
- Ordinal data OR normality assumption violated
- Small sample size with non-normal distribution

SCENARIO EXAMPLES:
- Comparing Likert scale ratings between groups
- Skewed data (e.g., income, reaction times)
- Small samples that fail normality test

ASSUMPTIONS:
✓ Ordinal or continuous dependent variable
✓ Two independent groups

✓ Independent observations
× Does NOT assume normality
× Does NOT assume equal variances
~ Assumes similar shape of distributions (for median test)

METHOD:
1. Rank all observations from both groups together
2. Sum ranks for each group (R_1 and R_2)
3. Calculate U statistic

FORMULAS:
$U_1 = n_1 \times n_2 + n_1(n_1+1)/2 - R_1$
$U_2 = n_1 \times n_2 + n_2(n_2+1)/2 - R_2$

Test statistic: $U = \min(U_1, U_2)$

For large samples (n_1, $n_2 > 20$), use normal approximation:
$z = (U - \mu_U) / \sigma_U$

Where:
$\mu_U = n_1 \times n_2 / 2$
$\sigma_U = \sqrt{(n_1 \times n_2 \times (n_1+n_2+1) / 12)}$

R CODE:
```
wilcox.test(DV ~ Group, data=data)
```

INTERPRETATION:
Tests whether one group tends to have larger values than the other
p-value < α → Distributions differ

EFFECT SIZE (r):
$r = Z / \sqrt{N}$

Where: N = total sample size = $n_1 + n_2$
Interpretation: Small: 0.1, Medium: 0.3, Large: 0.5

5.7 WILCOXON SIGNED-RANK TEST

USE WHEN:
- Comparing TWO RELATED groups
- Non-parametric alternative to paired t-test
- Ordinal data OR normality of differences violated

SCENARIO EXAMPLES:
- Before vs after with non-normal differences
- Matched pairs with ordinal outcomes

• Small paired samples

ASSUMPTIONS:
✓ Ordinal or continuous dependent variable
✓ Paired observations
✓ Differences are symmetric around median
× Does NOT assume normality

METHOD:
1. Calculate differences: $d_i = x_{1i} - x_{2i}$
2. Remove zeros (ties at difference = 0)
3. Rank absolute values of differences: $|d_i|$
4. Assign signs back to ranks
5. Sum positive ranks (W^+) and negative ranks (W^-)

FORMULA:
Test statistic: W = smaller of W^+ or W^-

R CODE:
```
wilcox.test(x1, x2, paired=TRUE)
```

INTERPRETATION:
Tests whether median difference differs from zero
p-value < α → Distributions differ

5.8 EFFECT SIZE: COHEN'S d

PURPOSE: Standardized measure of effect size (magnitude of difference)
• Independent of sample size
• Allows comparison across studies
• Complements significance testing

FORMULAS:

Independent samples:
$d = (M_1 - M_2) / SD_{pooled}$

Paired samples:
$d = M_{diff} / SD_{diff}$

One sample:
$d = (M - \mu_0) / SD$

INTERPRETATION (Cohen's conventions):
• Small effect: d = 0.2 (subtle difference)
• Medium effect: d = 0.5 (noticeable difference)

• Large effect: d = 0.8 (obvious difference)

RELATIONSHIP TO OVERLAP:
d = 0.0 → 100% overlap between distributions
d = 0.5 → 67% overlap
d = 0.8 → 53% overlap
d = 2.0 → 14% overlap

R CODE:
```
library(effsize)
cohen.d(DV ~ Group, data=data)
```

WHY REPORT EFFECT SIZE:
• Significance depends on sample size (large n → easier to get p<.05)
• Effect size tells you if the difference is MEANINGFUL
• Required by many journals
• Needed for power analysis and meta-analysis

5.9 CHECKING NORMALITY: SHAPIRO-WILK TEST

PURPOSE: Test whether data come from normal distribution

USE WHEN:
• Checking assumptions for parametric tests
• Small to medium sample sizes (n < 50 is most informative)
• Want formal statistical test of normality

HYPOTHESES:
H_0: Data are normally distributed
H_1: Data are not normally distributed

FORMULA:
$$W = (\Sigma a_i \times x_{(i)})^2 / \Sigma(x_i - \bar{x})^2$$

Where:
• $x_{(i)}$ = ordered sample values
• a_i = coefficients based on expected normal order statistics

INTERPRETATION:
W ranges from 0 to 1
• W close to 1 → Data consistent with normality
• W far from 1 → Data deviate from normality

R CODE:
```
shapiro.test(x)
```

DECISION:
p-value < α → Reject H_0, data are NOT normal
p-value ≥ α → Fail to reject, data consistent with normality

LIMITATIONS:
• Sensitive to sample size
  - Small n: Low power, may miss non-normality
  - Large n: High power, detects trivial deviations
• Always complement with visual inspection (Q-Q plots, histograms)

GUIDELINES:
• If p > .05 AND Q-Q plot looks good → assume normality
• If p < .05 BUT n > 30 and Q-Q plot acceptable → parametric tests may be robust
• If p < .05 AND Q-Q plot bad → use non-parametric tests or transform data

ALTERNATIVES:
• Kolmogorov-Smirnov test (less powerful)
• Anderson-Darling test (more powerful)
• Visual inspection (often sufficient for large n)

SUMMARY TABLE: TESTS FOR COMPARING MEANS

Test	Groups	Design	DV Type	Key Assumption	Alternative
One-sample z-test	1	Single	Continuous	σ known	None
One-sample t-test	1	Single	Continuous	σ unknown	Wilcoxon
Independent t-test	2	Between	Continuous	Equal variances	Mann-Whitney
Welch's t-test	2	Between	Continuous	Unequal variances	Mann-Whitney
Paired t-test	2	Within	Continuous	Normal differences	Wilcoxon SR
Mann-Whitney	2	Between	Ordinal	None (non-param)	-
Wilcoxon SR	2	Within	Ordinal	Symmetric diffs	-

---

6. CORRELATION AND REGRESSION

6.1 VARIANCE AND COVARIANCE

VARIANCE:
Measures spread of single variable around its mean

Population variance:
$\sigma^2 = \Sigma(x - \mu)^2 / N$

Sample variance (unbiased):

$s^2 = \Sigma(x - \bar{x})^2 / (n-1)$

Standard deviation: $\sigma = \sqrt{(\sigma^2)}$

COVARIANCE:
Measures how two variables vary together

Population covariance:
$Cov(X,Y) = \Sigma(x - \mu_X)(y - \mu_Y) / N$

Sample covariance:
$Cov(X,Y) = \Sigma(x - \bar{x})(y - \bar{y}) / (n-1)$

Properties:
- Cov > 0: Variables move together (positive relationship)
- Cov < 0: Variables move oppositely (negative relationship)
- Cov = 0: No linear relationship

LIMITATION: Scale-dependent (units = units_X × units_Y)
Solution: Standardize → Correlation coefficient

6.2 PEARSON'S CORRELATION (r)

USE WHEN:
- Measuring LINEAR relationship between two continuous variables
- Both variables approximately normally distributed
- No extreme outliers

SCENARIO EXAMPLES:
- Relationship between height and weight
- Study time and exam score
- Age and reaction time

ASSUMPTIONS:
✓ Both variables continuous (interval/ratio)
✓ Linear relationship
✓ Bivariate normality
✓ No extreme outliers
✓ Homoscedasticity

FORMULA:
$r = Cov(X,Y) / (\sigma_X \times \sigma_Y)$

Or equivalently:
$r = \Sigma[(x-\bar{x})(y-\bar{y})] / \sqrt{[\Sigma(x-\bar{x})^2 \times \Sigma(y-\bar{y})^2]}$

Or in terms of z-scores:

$r = \Sigma(z_x \times z_Y) / (n-1)$

PROPERTIES:

- Range: $-1 \leqslant r \leqslant 1$
- $r = +1$: Perfect positive linear relationship
- $r = -1$: Perfect negative linear relationship
- $r = 0$: No linear relationship

INTERPRETATION:

$|r| = 0.0-0.1$: Negligible

$|r| = 0.1-0.3$: Small/weak

$|r| = 0.3-0.5$: Medium/moderate

$|r| = 0.5-0.7$: Large/strong

$|r| = 0.7-0.9$: Very strong

$|r| = 0.9-1.0$: Extremely strong

HYPOTHESIS TEST:

H_0: $\rho = 0$ (no correlation in population)

H_1: $\rho \neq 0$

Test statistic:

$t = r\sqrt{(n-2)} / \sqrt{(1-r^2)}$

Degrees of freedom: $df = n - 2$

R CODE:

```
cor.test(x, y, method="pearson")
```

EFFECT SIZE:

r itself is the effect size

r^2 = coefficient of determination (proportion of variance explained)

IMPORTANT:

- Correlation ≠ Causation
- r measures only LINEAR relationships
- Outliers can heavily influence r
- Restricted range reduces r

6.3 SPEARMAN'S RANK CORRELATION (ρ)

USE WHEN:

- Measuring MONOTONIC relationship (not necessarily linear)
- Ordinal data
- Non-normal distributions
- Presence of outliers

• Non-parametric alternative to Pearson

SCENARIO EXAMPLES:
• Rank correlations (class rank vs test rank)
• Likert scale data
• Skewed distributions

ASSUMPTIONS:
✓ Ordinal, interval, or ratio data
✓ Monotonic relationship (not necessarily linear)
× Does NOT assume normality
× Does NOT assume linearity

METHOD:
1. Rank each variable separately
2. Apply Pearson's formula to ranks

FORMULA:
$\rho = 1 - (6\Sigma d^2) / (n(n^2-1))$

Where: d = difference between ranks for each pair

INTERPRETATION:
Same as Pearson's r: -1 to +1

R CODE:
```
cor.test(x, y, method="spearman")
```

WHEN TO CHOOSE SPEARMAN vs PEARSON:
• Linear relationship + normal data → Pearson (more powerful)
• Monotonic but not linear → Spearman
• Ordinal data → Spearman
• Outliers present → Spearman (more robust)

6.4 KENDALL'S TAU (τ)

USE WHEN:
• Small sample size
• Many tied ranks
• Alternative to Spearman
• More robust to outliers than Spearman

INTERPRETATION:
Similar to Spearman, but based on concordant/discordant pairs
Range: -1 to +1

METHOD:
Counts pairs that are in same order (concordant) vs opposite order (discordant)

FORMULA:
$\tau = (n_{concordant} - n_{discordant}) / (n(n-1)/2)$

R CODE:
```
cor.test(x, y, method="kendall")
```

COMPARISON TO SPEARMAN:
- Generally lower absolute values than Spearman
- More robust to ties
- Better for small samples ($n < 30$)
- Slower to compute for large samples

6.5 TESTING CORRELATIONS

SINGLE CORRELATION:
$H_0: \rho = 0$

Test statistic:
$t = r\sqrt{(n-2)} / \sqrt{(1-r^2)}$
$df = n - 2$

COMPARING TWO INDEPENDENT CORRELATIONS:
$H_0: \rho_1 = \rho_2$ (correlations from different samples are equal)

Fisher's r-to-z transformation:
$z = 0.5 \times \ln[(1+r)/(1-r)]$

Test statistic:
$Z = (z_1 - z_2) / \sqrt{(1/(n_1-3) + 1/(n_2-3))}$

COMPARING TWO DEPENDENT CORRELATIONS (SAME SAMPLE):
$H_0: \rho_{12} = \rho_{13}$ (correlations with same variable)

Example: Is correlation between X and Y different from correlation between X and Z?

Use specialized tests (Steiger's Z, Williams' t)

R CODE:
```
library(cocor)
cocor.dep.groups.overlap(r.jk, r.jh, r.kh, n)
```

6.6 SIMPLE LINEAR REGRESSION

USE WHEN:

- Predicting continuous outcome (Y) from one predictor (X)
- Modeling linear relationship
- Want to quantify relationship strength

SCENARIO EXAMPLES:

- Predicting salary from years of experience
- Predicting exam score from study hours
- Predicting weight from height

ASSUMPTIONS:

✓ Linearity: Relationship between X and Y is linear
✓ Independence: Observations are independent
✓ Normality: Residuals are normally distributed
✓ Homoscedasticity: Constant variance of residuals
✓ No influential outliers

MODEL:

$Y = \beta_0 + \beta_1 X + \varepsilon$

Where:

- Y = dependent variable (outcome)
- X = independent variable (predictor)
- β_0 = intercept (value of Y when X=0)
- β_1 = slope (change in Y per unit change in X)
- ε = error term (residual)

ESTIMATION (Ordinary Least Squares):

Minimize sum of squared residuals: $\Sigma(y_i - \hat{y}_i)^2$

Slope:

$\beta_1 = \Sigma[(x-\bar{x})(y-\bar{y})] / \Sigma(x-\bar{x})^2$
$= Cov(X,Y) / Var(X)$
$= r \times (SD_Y / SD_X)$

Intercept:

$\beta_0 = \bar{y} - \beta_1 \bar{x}$

PREDICTION:

$\hat{y} = \beta_0 + \beta_1 x$

RESIDUAL:

$e_i = y_i - \hat{y}_i$

R CODE:

```
model <- lm(Y ~ X, data=data)
summary(model)
```

OUTPUT INTERPRETATION:

- Coefficients: β_0 and β_1 estimates
- Std. Error: Uncertainty in coefficients
- t value: Coefficient / Std. Error
- Pr(>|t|): p-value for testing H_0: $\beta = 0$
- R^2: Proportion of variance explained
- F-statistic: Overall model significance

HYPOTHESIS TESTS:

For slope:

H_0: $\beta_1 = 0$ (no relationship)

Test statistic:

$t = \beta_1 / SE(\beta_1)$

$df = n - 2$

Confidence interval for slope:

$CI = \beta_1 \pm t_critical \times SE(\beta_1)$

6.7 MULTIPLE LINEAR REGRESSION

USE WHEN:

- Predicting continuous outcome from MULTIPLE predictors
- Want to control for confounds
- Test unique contribution of each predictor

SCENARIO EXAMPLES:

- Predicting salary from education, experience, and age
- Predicting exam score from study hours, sleep, and motivation
- Predicting reaction time from age, caffeine, and practice

ASSUMPTIONS:

✓ Linearity
✓ Independence of observations
✓ Normality of residuals
✓ Homoscedasticity
✓ No multicollinearity (predictors not too correlated)
✓ No influential outliers

MODEL:

$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_k X_k + \varepsilon$

Where:

- Y = outcome
- $X_1, X_2, \ldots, X_k$ = predictors
- β_0 = intercept
- $\beta_1, \beta_2, \ldots, \beta_k$ = partial regression coefficients
- ε = error term

INTERPRETATION OF COEFFICIENTS:
β_i = change in Y per unit change in X_i, HOLDING ALL OTHER PREDICTORS CONSTANT

Example:
$Y = 50 + 2X_1 + 3X_2$
- $\beta_1 = 2$: Increasing X_1 by 1 (keeping X_2 constant) increases Y by 2
- $\beta_2 = 3$: Increasing X_2 by 1 (keeping X_1 constant) increases Y by 3

R CODE:
```
model <- lm(Y ~ X1 + X2 + X3, data=data)
summary(model)
```

HYPOTHESIS TESTS:
For each coefficient:
H_0: $\beta_i = 0$ (predictor has no unique effect)

Test statistic:
$t = \beta_i / SE(\beta_i)$
$df = n - k - 1$

Where: k = number of predictors

For overall model:
H_0: $\beta_1 = \beta_2 = \ldots = \beta_k = 0$ (model has no predictive value)

Test statistic:
$F = (R^2 / k) / ((1-R^2) / (n-k-1))$

6.8 MODEL FIT: R^2 and Adjusted R^2

R^2 (Coefficient of Determination):
$R^2 = 1 - (SS_residual / SS_total)$
$\quad = SS_regression / SS_total$

Where:
- $SS_total = \Sigma(y - \bar{y})^2$ (total variance)
- $SS_regression = \Sigma(\hat{y} - \bar{y})^2$ (explained variance)
- $SS_residual = \Sigma(y - \hat{y})^2$ (unexplained variance)

INTERPRETATION:
Proportion of variance in Y explained by the model
• $R^2 = 0$: Model explains no variance (useless)
• $R^2 = 1$: Model explains all variance (perfect fit)
• $R^2 = 0.30$: Model explains 30% of variance

PROBLEM WITH R^2:
Always increases when adding predictors, even if they're useless!

ADJUSTED R^2 (Penalizes Model Complexity):
$R^2_{adj} = 1 - [(1-R^2)(n-1) / (n-k-1)]$

Where:
• n = sample size
• k = number of predictors

INTERPRETATION:
• Adjusts R^2 for number of predictors
• Can decrease if adding useless predictors
• Use for comparing models with different numbers of predictors

R CODE:
```
summary(model)$r.squared
summary(model)$adj.r.squared
```

RSS (Residual Sum of Squares):
$RSS = \Sigma(y - \hat{y})^2$

Lower RSS = better fit
Used in model comparison (F-tests)

RMSE (Root Mean Squared Error):
$RMSE = \sqrt{(RSS / n)}$

Interpretation: Average prediction error (in original units of Y)

SUMMARY: WHEN TO USE WHICH CORRELATION/REGRESSION

Scenario	Method
Linear relationship, normal data	Pearson's r
Monotonic relationship, ordinal	Spearman's ρ
Small sample, ties	Kendall's τ
One predictor (continuous)	Simple linear regression
Multiple predictors	Multiple linear regression
Binary outcome	Logistic regression (see GLMER)

Count outcome Poisson regression (see GLMER)

==========

7. REGRESSION DIAGNOSTICS AND SIGNIFICANCE TESTING

==========

7.1 MODEL COMPARISON (F-TEST)

PURPOSE: Compare nested models to test if additional predictors improve fit

USE WHEN:

- Testing whether adding predictor(s) significantly improves model
- Comparing simple model vs complex model
- Models must be NESTED (simpler model is subset of complex model)

SCENARIO EXAMPLES:

- Model 1: Y ~ X1
  Model 2: Y ~ X1 + X2
  Question: Does X2 significantly improve prediction?

- Model 1: Y ~ X1 + X2
  Model 2: Y ~ X1 + X2 + X1:X2
  Question: Is the interaction significant?

HYPOTHESES:

H_0: Reduced model is sufficient (extra predictors don't help)

H_1: Full model is significantly better

FORMULA:

F = [(RSS_reduced - RSS_full) / (df_reduced - df_full)] / [RSS_full / df_full]

Or equivalently:

F = [(R^2_full - R^2_reduced) / (k_full - k_reduced)] / [(1-R^2_full) / (n-k_full-1)]

Degrees of freedom:

- df_1 = k_full - k_reduced (number of additional predictors)
- df_2 = n - k_full - 1

R CODE:

```
model1 <- lm(Y ~ X1, data=data)
model2 <- lm(Y ~ X1 + X2, data=data)
anova(model1, model2)
```

OUTPUT INTERPRETATION:

- RSS: Residual sum of squares for each model
- Df: Degrees of freedom
- F: Test statistic

• Pr(>F): p-value

DECISION:
p < α → Full model is significantly better, keep additional predictor(s)
p ≥ α → Reduced model is sufficient, extra predictor(s) not needed

IMPORTANT:
• Only for NESTED models
• Models must be fitted on SAME data
• For comparing mixed models: Use ML (not REML) when comparing fixed effects

---

7.2 SEQUENTIAL F-TESTS (ANOVA FOR REGRESSION)

PURPOSE: Test contribution of each predictor when added sequentially

METHOD:
Build up model step by step, test improvement at each step

Example:
Model 0: Y ~ 1 (intercept only)
Model 1: Y ~ X1
Model 2: Y ~ X1 + X2
Model 3: Y ~ X1 + X2 + X3

Test each step with F-test

R CODE:
```
model <- lm(Y ~ X1 + X2 + X3, data=data)
anova(model)  # Sequential ANOVA
```

OUTPUT:
Shows contribution of each predictor IN THE ORDER ENTERED
• Order matters! Results depend on order of predictors

TYPE I SS (Sequential):
Each predictor tested after all previous predictors
Order-dependent

TYPE II SS (Partial):
Each predictor tested after all others (no interactions)
Order-independent

TYPE III SS (Marginal):
Each predictor tested in presence of all others (including interactions)
Order-independent, most common in ANOVA

```
R CODE FOR TYPE II/III:
library(car)
Anova(model, type="II")
Anova(model, type="III")

--------------------------------------------------------------------------------

7.3 TESTING REGRESSION ASSUMPTIONS
--------------------------------------------------------------------------------

ASSUMPTION 1: LINEARITY
What: Relationship between X and Y is linear

Check:
• Scatterplot of Y vs X
• Residual plot: Residuals vs Fitted values
  - Should show random scatter around zero
  - Pattern indicates non-linearity

Remedy:
• Transform variables (log, sqrt, polynomial)
• Add polynomial terms: X + X²
• Use non-linear models

--------------------------------------------------------------------------------

ASSUMPTION 2: INDEPENDENCE
What: Observations are independent (one doesn't influence another)

Check:
• Consider study design
• Durbin-Watson test (for time series)
• Plot residuals vs observation order

Violation examples:
• Repeated measures from same subjects
• Time series data (autocorrelation)
• Clustered data (students within schools)

Remedy:
• Use mixed-effects models
• Add time/cluster as random effect
• Use generalized estimating equations (GEE)

--------------------------------------------------------------------------------

ASSUMPTION 3: NORMALITY OF RESIDUALS
What: Residuals are normally distributed
```

Check:
• Q-Q plot of residuals
  - Points should fall on diagonal line
• Shapiro-Wilk test on residuals
• Histogram of residuals

R CODE:
```
plot(model, which=2)  # Q-Q plot
shapiro.test(residuals(model))
```

Important: Test normality of RESIDUALS, not raw data!

Remedy:
• Transform Y (log, sqrt, inverse)
• Use robust regression
• Use bootstrap methods
• If n large, less critical (CLT)

ASSUMPTION 4: HOMOSCEDASTICITY (EQUAL VARIANCE)
What: Variance of residuals is constant across fitted values

Check:
• Residuals vs Fitted plot
  - Should show constant spread (no funnel shape)
• Scale-Location plot (sqrt of standardized residuals vs fitted)
  - Horizontal red line indicates homoscedasticity
• Breusch-Pagan test or White test

R CODE:
```
plot(model, which=1)  # Residuals vs Fitted
plot(model, which=3)  # Scale-Location

library(lmtest)
bptest(model)  # Breusch-Pagan test
```

Violation: Heteroscedasticity
• Funnel shape in residual plot
• Variance increases/decreases with fitted values

Remedy:
• Transform Y (log, sqrt)
• Use weighted least squares (WLS)
• Use robust standard errors
• Use generalized linear models (GLM)

```
--------------------------------------------------------------------------------

7.4 OUTLIERS, LEVERAGE, AND COOK'S DISTANCE
--------------------------------------------------------------------------------

OUTLIERS:
Observations with large residuals (far from regression line)

Detection:
• Standardized residuals > |2| or |3|
• Studentized residuals > |3|

R CODE:
rstandard(model)  # Standardized residuals
rstudent(model)   # Studentized residuals

Effect: Increase error variance, reduce R²

--------------------------------------------------------------------------------

LEVERAGE:
Observations with extreme predictor values (far from X mean)

Measure: Hat values (diagonal of hat matrix)

hᵢᵢ = leverage for observation i

Threshold: h > 2(k+1)/n or h > 3(k+1)/n

Where: k = number of predictors

Detection:
plot(model, which=5)  # Residuals vs Leverage

Effect: Potential to influence regression line (high leverage = potential influence)

--------------------------------------------------------------------------------

COOK'S DISTANCE:
Measures INFLUENCE = combines outlier + leverage

Formula:
D_i = (e_i²/k+1) × (h_i/(1-h_i)²)

Where:
• e_i = standardized residual
• h_i = leverage
• k = number of predictors
```

Interpretation:
Measures how much regression coefficients would change if observation removed

Threshold:
• D > 1: Definitely influential
• D > 4/n: Common threshold
• D > 4/(n-k-1): Conservative threshold

Detection:
R CODE:
```
plot(model, which=4)  # Cook's distance
plot(model, which=5)  # Residuals vs Leverage with Cook's

cooks.distance(model)
```

DECISION:
If D > threshold:
1. Check for data entry errors
2. Investigate why observation is unusual
3. Run analysis with and without observation
4. Report sensitivity analysis
5. Consider if it represents important subpopulation

DON'T automatically remove influential points!

7.5 MULTICOLLINEARITY

PROBLEM: Predictors are highly correlated with each other

Effects:
• Inflated standard errors → less precise estimates
• Unstable coefficients → small data changes cause large estimate changes
• Difficulty interpreting individual effects
• Reduced power to detect effects

Detection Methods:

1. Correlation Matrix:
Correlations among predictors
|r| > 0.7 or 0.8 is concerning

R CODE:
```
cor(data[, c("X1", "X2", "X3")])
```

2. Variance Inflation Factor (VIF):
Measures how much variance of coefficient is inflated due to multicollinearity

Formula:
$VIF_j = 1 / (1 - R^2_j)$

Where: R^2_j from regressing X_j on all other predictors

Interpretation:
- VIF = 1: No correlation with other predictors
- VIF = 5-10: Moderate multicollinearity (concerning)
- VIF > 10: Severe multicollinearity (problematic)

VIF tells you: SE is inflated by factor of $\sqrt{VIF}$

Example: VIF = 4 means SE is 2× larger than if predictors uncorrelated

R CODE:
```
library(car)
vif(model)
```

3. Tolerance:
Tolerance = 1 / VIF

Tolerance < 0.1 or 0.2 indicates multicollinearity

4. Condition Index:
From eigenvalue decomposition
CI > 30 indicates multicollinearity

Remedies:
1. Remove one of the correlated predictors
2. Combine correlated predictors (average, PCA)
3. Center predictors (especially for interactions)
4. Increase sample size
5. Use ridge regression or regularization

IMPORTANT:
Multicollinearity affects INTERPRETATION and PRECISION, not PREDICTION
If only predicting, multicollinearity less problematic

DIAGNOSTIC PLOTS IN R

```
plot(model)  # Shows 4 diagnostic plots
```

Plot 1: Residuals vs Fitted

• Check: Linearity, homoscedasticity
• Look for: Random scatter around zero
• Red line should be flat and close to zero

Plot 2: Q-Q Plot
• Check: Normality of residuals
• Look for: Points on diagonal line
• Deviations at tails indicate non-normality

Plot 3: Scale-Location (Spread-Location)
• Check: Homoscedasticity
• Look for: Horizontal red line
• Upward/downward trend indicates heteroscedasticity

Plot 4: Residuals vs Leverage
• Check: Influential observations
• Look for: Points outside Cook's distance contours
• Labels show influential observations

Plot 5: (with which=5)
• Shows Cook's distance contours
• Identifies influential cases

==========
8. CATEGORICAL PREDICTORS AND INTERACTIONS IN REGRESSION
==========

8.1 FACTORIAL ANOVA CONCEPTS

FACTORIAL DESIGN:
Multiple independent variables (factors) with multiple levels

Example: 2×3 design
• Factor A: 2 levels (e.g., Gender: Male, Female)
• Factor B: 3 levels (e.g., Treatment: Control, Drug1, Drug2)
• Total: 2 × 3 = 6 conditions

EFFECTS TO TEST:
1. Main effect of Factor A
2. Main effect of Factor B
3. Interaction between A and B

MAIN EFFECT:
Overall effect of one factor, averaged across levels of other factor(s)

INTERACTION:
Effect of one factor DEPENDS ON level of another factor

• Lines are not parallel in interaction plot
• Effect of A differs at different levels of B

INTERPRETING INTERACTIONS:
1. If interaction significant → interpret interaction, not main effects alone
2. Simple effects analysis: Test effect of A at each level of B
3. Plot interaction to visualize pattern

R CODE (as regression):
```
model <- lm(Y ~ FactorA * FactorB, data=data)
```

R CODE (as ANOVA):
```
model <- aov(Y ~ FactorA * FactorB, data=data)
summary(model)
```

ASSUMPTIONS (same as regression):
✓ Normality
✓ Homogeneity of variance
✓ Independence

8.2 CODING SCHEMES FOR CATEGORICAL PREDICTORS

When categorical predictor has k levels, create k-1 dummy variables

DUMMY CODING (Treatment Coding):
One level as reference (baseline), others compared to it

Example: 3 levels (Control, Drug1, Drug2)
Choose Control as reference:

```
         Drug1  Drug2
Control    0      0     ← Reference
Drug1      1      0
Drug2      0      1
```

Interpretation:
• Intercept = Mean of reference group (Control)
• β_1 = Difference between Drug1 and Control
• β_2 = Difference between Drug2 and Control

R CODE:
```
# R uses treatment coding by default
# Reference = first level alphabetically
# To change reference:
data$treatment <- relevel(data$treatment, ref="Control")
```

```
model <- lm(Y ~ treatment, data=data)

COEFFICIENTS:
(Intercept)        = Mean_Control
treatmentDrug1     = Mean_Drug1 - Mean_Control
treatmentDrug2     = Mean_Drug2 - Mean_Control

--------------------------------------------------------------------------------

SUM CODING (Effects Coding):
Compares each level to grand mean

Example: 3 levels (Control, Drug1, Drug2)

         D1   D2
Control  -1   -1     ← Sum to zero
Drug1     1    0
Drug2     0    1

Interpretation:
• Intercept = Grand mean (average of all group means)
• β₁ = Difference between Drug1 and grand mean
• β₂ = Difference between Drug2 and grand mean
• Third level (Control) coefficient = -(β₁ + β₂)

R CODE:
contrasts(data$treatment) <- contr.sum(3)
model <- lm(Y ~ treatment, data=data)

WHEN TO USE WHICH:
• Treatment coding: When one group is natural baseline (control, placebo)
• Sum coding: When all groups are equally important, no natural reference
• Results are equivalent (same F-tests), only interpretation differs

--------------------------------------------------------------------------------

8.3 INTERACTIONS BETWEEN CATEGORICAL AND CONTINUOUS PREDICTORS
--------------------------------------------------------------------------------

MODEL:
Y = β₀ + β₁×Group + β₂×X + β₃×(Group×X) + ε

Where:
• Group = categorical (0/1 with dummy coding)
• X = continuous predictor
• Group×X = interaction term
```

INTERPRETATION:

Without interaction: $Y = \beta_0 + \beta_1 \times Group + \beta_2 \times X$
- Parallel lines for each group
- Same slope (β_2) for all groups
- Different intercepts (β_0 and $\beta_0+\beta_1$)

With interaction: $Y = \beta_0 + \beta_1 \times Group + \beta_2 \times X + \beta_3 \times (Group \times X)$
- Non-parallel lines
- Different slopes for each group
- Different intercepts for each group

EQUATIONS FOR EACH GROUP:

Group 0 (reference): $Y = \beta_0 + \beta_2 \times X$
- Intercept: β_0
- Slope: β_2

Group 1: $Y = (\beta_0 + \beta_1) + (\beta_2 + \beta_3) \times X$
- Intercept: $\beta_0 + \beta_1$
- Slope: $\beta_2 + \beta_3$

HYPOTHESIS TEST FOR INTERACTION:
H_0: $\beta_3 = 0$ (slopes are equal, no interaction)
H_1: $\beta_3 \neq 0$ (slopes differ, interaction exists)

R CODE:
```
model <- lm(Y ~ Group * X, data=data)
# Equivalent to:
model <- lm(Y ~ Group + X + Group:X, data=data)
```

INTERPRETATION OF β_3:
"Difference in slopes between groups"
- $\beta_3 > 0$: Group 1 has steeper positive slope than Group 0
- $\beta_3 < 0$: Group 1 has less steep (or more negative) slope than Group 0

VISUALIZATION:
```
plot(X, Y, col=Group)
abline(a = β0, b = β2, col=1)              # Group 0
abline(a = β0+β1, b = β2+β3, col=2)        # Group 1
```

SIMPLE SLOPES ANALYSIS:
Test whether slope is significant in each group separately

For Group 0:
Test H_0: $\beta_2 = 0$

For Group 1:

Test H_0: $(\beta_2 + \beta_3) = 0$

R CODE:

```
library(interactions)
sim_slopes(model, pred=X, modx=Group)
```

REGIONS OF SIGNIFICANCE:
Johnson-Neyman technique: Identify values of continuous variable where group differences are significant

---

8.4 POST-HOC TESTING AND PLANNED CONTRASTS

When overall F-test is significant in ANOVA:
→ Need follow-up tests to determine which groups differ

TWO APPROACHES:

1. POST-HOC TESTS (Unplanned comparisons)
- Test all pairwise differences
- Control for multiple comparisons (family-wise error rate)

2. PLANNED CONTRASTS (A priori comparisons)
- Specific comparisons planned before seeing data
- Based on theoretical predictions
- Less conservative (more power)

COMMON POST-HOC TESTS:

Test	Use When	Controls	Conservativeness
Tukey HSD	All pairwise comparisons	FWER	Moderate
Bonferroni	Few planned comparisons	FWER	Very conservative
Holm	Sequential Bonferroni	FWER	Less conservative
Scheffé	Any/all possible contrasts	FWER	Most conservative
Dunnett	Compare all to control	FWER	Moderate

BONFERRONI CORRECTION:
Adjusted $\alpha = \alpha / m$

Where: m = number of comparisons

Example: 3 pairwise tests, $\alpha = .05$
Adjusted $\alpha = .05/3 = .0167$
Use $\alpha = .0167$ for each test

Pros: Simple, controls FWER
Cons: Conservative (loses power), treats tests as independent

R CODE:
```
pairwise.t.test(Y, Group, p.adjust.method="bonferroni")
```

TUKEY HSD (Honestly Significant Difference):
Optimal for all pairwise comparisons
Accounts for dependency among tests

R CODE:
```
model <- aov(Y ~ Group, data=data)
TukeyHSD(model)
```

PLANNED CONTRASTS:
Specify contrast weights that sum to zero

Example: Compare Control vs average of two treatments
Weights: Control=-2, Treatment1=1, Treatment2=1

R CODE:
```
contrasts(data$Group) <- cbind(
  c1 = c(-2, 1, 1)  # Control vs Treatments
)
model <- lm(Y ~ Group, data=data)
summary(model)
```

SIMPLE EFFECTS:
Test effect of one factor at each level of another factor

Example: Effect of Treatment at each Gender
• Effect of Treatment for Males
• Effect of Treatment for Females

R CODE:
```
library(emmeans)
emm <- emmeans(model, ~ Treatment | Gender)
pairs(emm)
```

8.5 EXPERIMENTAL DESIGNS

WITHIN-SUBJECT (REPEATED MEASURES / RELATED):
Same participants in all conditions

```
Examples:
• Before vs after treatment
• Practice trials over time
• Different conditions tested on same day

Advantages:
• Controls for individual differences
• More statistical power
• Fewer participants needed

Disadvantages:
• Order effects (practice, fatigue)
• Carryover effects
• May not be feasible (e.g., can't un-learn something)

Analysis:
• Paired t-test (2 conditions)
• Repeated measures ANOVA (3+ conditions)
• Mixed-effects models with random intercepts

R CODE:
# Repeated measures ANOVA
aov_model <- aov(Y ~ Condition + Error(Subject/Condition), data=data)

# Mixed model
library(lme4)
lmer_model <- lmer(Y ~ Condition + (1|Subject), data=data)

ASSUMPTIONS:
• Sphericity: Equal variances of differences between all pairs
  - Test: Mauchly's test
  - Correction: Greenhouse-Geisser or Huynh-Feldt

--------------------------------------------------------------------------------

BETWEEN-SUBJECT (INDEPENDENT / UNRELATED):
Different participants in each condition

Examples:
• Treatment vs control groups
• Males vs females
• Young vs old participants

Advantages:
• No order/carryover effects
• Simple to implement
• Each participant only measured once

```

Disadvantages:
• Individual differences add noise
• Less power
• Need more participants

Analysis:
• Independent t-test (2 groups)
• One-way ANOVA (3+ groups)
• Factorial ANOVA (multiple factors)

R CODE:
```
# Independent ANOVA
aov_model <- aov(Y ~ Group, data=data)

# As regression
lm_model <- lm(Y ~ Group, data=data)
```

MIXED DESIGNS:
Combination of within-subject and between-subject factors

Example:
• Factor A: Treatment (between-subjects: Group1 vs Group2)
• Factor B: Time (within-subjects: Pre vs Post)

Analysis:
Mixed ANOVA or Mixed-effects models

R CODE:
```
aov(Y ~ Treatment*Time + Error(Subject/Time), data=data)

lmer(Y ~ Treatment*Time + (Time|Subject), data=data)
```

==========
9. LINEAR MIXED-EFFECTS MODELS (LMER)
==========

9.1 WHEN TO USE LMER

USE WHEN:
• Non-independent data (repeated measures, clustered, hierarchical)
• Multiple observations per unit (subject, school, item, etc.)
• Want to generalize beyond specific sample (random effects)
• Unbalanced designs (different n per group)
• Missing data (handles better than repeated measures ANOVA)

EXAMPLES:

• Students (repeated) nested in schools
• Multiple trials per participant
• Participants rate multiple items (crossed design)
• Longitudinal data (time within person)

ADVANTAGES OVER TRADITIONAL ANOVA:
• Handles missing data
• Flexible correlation structures
• Continuous and categorical predictors
• Generalizability (random effects)
• Better for unbalanced designs

9.2 FIXED VS RANDOM EFFECTS

FIXED EFFECTS:
• Parameters we want to test/estimate
• Focus of research question
• Repeatable, systematic influences
• Assume same value for everyone
• Generalize to these specific levels only

Examples:
• Treatment (Drug A vs Drug B vs Control)
• Gender (Male vs Female)
• Time (specific time points)
• Age group

Question: "What is THE effect of X on Y?"

RANDOM EFFECTS:
• Sources of random variation
• Model nuisance variability
• Sample from larger population
• Different value for each unit
• Generalize beyond sampled levels

Examples:
• Participant ID (not interested in specific people)
• Item/stimulus ID (sample of all possible items)
• School ID (sample of all schools)
• Batch effects in lab studies

Question: "How much does the effect of X vary across units?"

KEY DISTINCTION:

Fixed: "What is the effect?"
Random: "How much does it vary?"

MATHEMATICAL REPRESENTATION:

Fixed effects only:
$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$

Mixed effects:
$Y_{ij} = (\beta_0 + u_{0j}) + (\beta_1 + u_{1j})X_{ij} + \varepsilon_{ij}$

Where:
- β_0 = population intercept (fixed)
- u_{0j} = deviation of unit j from population intercept (random)
- β_1 = population slope (fixed)
- u_{1j} = deviation of unit j's slope from population slope (random)
- ε_{ij} = residual

Random effects assumed: $u_{0j} \sim N(0, \sigma^2_u0)$
$u_{1j} \sim N(0, \sigma^2_u1)$

9.3 RANDOM INTERCEPTS

MODEL:
$Y_{ij} = (\beta_0 + u_{0j}) + \beta_1 X_{ij} + \varepsilon_{ij}$

Where:
- i = observation within unit j
- j = unit (subject, school, etc.)
- u_{0j} = random intercept for unit j

INTERPRETATION:
- Each unit has its own baseline (intercept)
- Units differ in overall level of Y
- Slopes are same for all units (parallel lines)

EXAMPLE:
Students nested in schools:
- Each school has different average test score (random intercept)
- Effect of study time is same for all schools (fixed slope)

R CODE:
```
model <- lmer(Y ~ X + (1|Subject), data=data)
```

NOTATION:

(1|Subject) means:
• 1 = intercept
• | = "varying by"
• Subject = grouping variable

READ AS: "Random intercept for Subject"

VARIANCE COMPONENTS:
Model estimates:
• σ^2_{u0} = variance of random intercepts (between-subject variance)
• σ^2_{ε} = residual variance (within-subject variance)

INTRACLASS CORRELATION (ICC):
$ICC = \sigma^2_{u0} / (\sigma^2_{u0} + \sigma^2_{\varepsilon})$

Interpretation:
• Proportion of total variance due to grouping
• ICC = 0: No clustering (random effects unnecessary)
• ICC = 1: All variance between groups, none within
• ICC = 0.3: 30% of variance due to group membership

RULE OF THUMB:
ICC > 0.05 → Random effects needed

9.4 RANDOM SLOPES

MODEL:
$Y_{ij} = (\beta_0 + u_{0j}) + (\beta_1 + u_{1j})X_{ij} + \varepsilon_{ij}$

Where:
• u_{0j} = random intercept for unit j
• u_{1j} = random slope for unit j

INTERPRETATION:
• Each unit has its own intercept AND slope
• Effect of X varies across units
• Non-parallel lines

EXAMPLE:
• Each student has different baseline (random intercept)
• Effect of study time differs for each student (random slope)
• Some students benefit more from study time than others

R CODE:
```
model <- lmer(Y ~ X + (X|Subject), data=data)
```

NOTATION:
(X|Subject) means:
• Random intercept for Subject
• Random slope for X by Subject
• Correlation between random intercepts and slopes

EQUIVALENT TO:
(1 + X|Subject)

VARIANCE COMPONENTS:
Model estimates:
• σ^2_{u0} = variance of random intercepts
• σ^2_{u1} = variance of random slopes
• ρ = correlation between random intercepts and slopes
• σ^2_{ε} = residual variance

CORRELATION INTERPRETATION:
• $\rho > 0$: Higher baseline → steeper slope (amplification)
• $\rho < 0$: Higher baseline → flatter slope (compensation)
• $\rho = 0$: Baseline and slope independent

UNCORRELATED RANDOM EFFECTS:
If you want to force correlation = 0:

R CODE:
```
model <- lmer(Y ~ X + (1|Subject) + (0+X|Subject), data=data)
```

Or using || syntax:
```
model <- lmer(Y ~ X + (X||Subject), data=data)
```

9.5 CROSSED VS NESTED RANDOM EFFECTS

NESTED RANDOM EFFECTS:
Lower-level units belong to one higher-level unit only

EXAMPLES:
• Students nested in schools
  - Student 1 is in School A only
• Measurements nested within subjects
  - Trial 1 for Subject 1 is unique to that subject

STRUCTURE:
School A: Student 1, Student 2, Student 3
School B: Student 4, Student 5, Student 6

```
→ Students are NESTED in schools

R CODE:
model <- lmer(Y ~ X + (1|School/Student), data=data)
# Equivalent to:
model <- lmer(Y ~ X + (1|School) + (1|School:Student), data=data)

NOTATION:
(1|School/Student) means:
• Random intercept for School
• Random intercept for Student within School

--------------------------------------------------------------------------------

CROSSED RANDOM EFFECTS:
Levels of one random factor occur across levels of another

EXAMPLES:
• Participants × Items design
  - Each participant sees each item
  - Each item is seen by each participant
• Raters × Targets
  - Each rater rates each target

STRUCTURE:
         Item1  Item2  Item3
Subj1     X      X      X
Subj2     X      X      X
Subj3     X      X      X
→ Subjects and Items are CROSSED

R CODE:
model <- lmer(Y ~ X + (1|Subject) + (1|Item), data=data)

NOTATION:
(1|Subject) + (1|Item) means:
• Random intercept for Subject
• Random intercept for Item
• No nesting implied

COMPLEX CROSSED DESIGNS:
With random slopes:

model <- lmer(Y ~ Condition +
              (Condition|Subject) +    # Subject varies in Condition response
              (1|Item),                # Item baseline only
              data=data)

```

WHY CROSSED MATTERS:
• Allows generalization to new subjects AND new items
• Accounts for both sources of variability
• Essential for stimulus-based research (psycholinguistics, judgment studies)

DECISION RULE:
• Can a level of factor A appear with multiple levels of factor B?
  - YES → Crossed
  - NO → Nested

9.6 MODEL COMPARISON: AIC, BIC

PURPOSE:
Compare non-nested models or models with different fixed effects

AKAIKE INFORMATION CRITERION (AIC):
AIC = -2×log(Likelihood) + 2k

Where: k = number of parameters

BAYESIAN INFORMATION CRITERION (BIC):
BIC = -2×log(Likelihood) + k×log(n)

Where: n = number of observations

INTERPRETATION:
• Lower = better
• Penalize model complexity
• BIC penalizes complexity more (especially large n)

DIFFERENCES:
• ΔAIC < 2: Models essentially equivalent
• ΔAIC = 2-4: Some support for lower AIC model
• ΔAIC = 4-7: Considerable support
• ΔAIC > 10: Strong support

R CODE:
AIC(model1, model2)
BIC(model1, model2)

WHEN TO USE:
• Comparing different fixed effects structures (fit with ML)
• Comparing different random effects structures (fit with REML or ML)
• Non-nested models

```
LIKELIHOOD RATIO TEST (LRT):
For NESTED models, more powerful than AIC/BIC

R CODE:
anova(model1, model2)

IMPORTANT:
• For fixed effects comparison: Use ML, not REML
• For random effects comparison: Can use REML

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9.7 MAXIMAL RANDOM EFFECTS STRUCTURE
--------------------------------------------------------------------------------

PRINCIPLE (Barr et al., 2013):
"Keep it maximal" - include all random effects justified by design

MAXIMAL MODEL:
Include random slopes for all within-unit factors

RULES:
1. Random intercepts for all grouping factors
2. Random slopes for all within-unit manipulated factors
3. Correlations between random effects

EXAMPLE:
Design: Subjects read items in two conditions (easy vs hard)
• Condition is within-subjects (each person sees both)
• Condition is within-items (each item appears in both)

MAXIMAL MODEL:
model <- lmer(RT ~ Condition +
              (Condition|Subject) +   # Slope: Condition varies within subjects
              (Condition|Item),       # Slope: Condition varies within items
              data=data)

Why maximal?
• Controls for individual differences in effects
• Reduces Type I error
• Appropriate generalization

WHEN TO SIMPLIFY:
If maximal model fails to converge:

1. Remove correlations:
model <- lmer(RT ~ Condition +
              (Condition||Subject) +   # No correlation
```

```
              (Condition||Item),
              data=data)

2. Remove random slopes with smallest variance:
model <- lmer(RT ~ Condition +
              (1|Subject) +            # Intercept only
              (Condition|Item),
              data=data)

3. Remove interaction random slopes (keep main effects):
model <- lmer(RT ~ A*B +
              (A+B|Subject),           # No A:B random slope
              data=data)

CONVERGENCE ISSUES:
Signs of problems:
• Singular fit warning
• Random effect variance = 0
• Random effect correlation = ±1
• Message about rescaling variables

Solutions:
• Simplify random effects (above strategies)
• Center/scale predictors
• Increase iterations: control=lmerControl(optCtrl=list(maxfun=100000))
• Try different optimizer: optimizer="bobyqa"

BALANCING ACT:
• Maximal → Best error control but convergence issues
• Simplified → Convergence but increased Type I error
• Document decisions, report attempts

--------------------------------------------------------------------------------

9.8 REPORTING LMER RESULTS
--------------------------------------------------------------------------------

MINIMUM TO REPORT:
1. Fixed effects estimates, SEs, t/z values, p-values
2. Random effects variances and SDs
3. Model formula in R syntax
4. Sample size and grouping levels (n subjects, n items, etc.)
5. Whether ML or REML used
6. Software and package versions

EXAMPLE:
"We fitted a linear mixed-effects model using lme4 (v1.1-35) in R (v4.3.1).
The model predicted reaction time from condition (easy vs hard) and included
```

```
random intercepts for subjects and items, as well as random slopes for
condition by subject: RT ~ Condition + (Condition|Subject) + (1|Item).
The model was fitted using REML.

Fixed effects showed a significant effect of condition (β = 45.2ms, SE = 8.3,
t = 5.45, p < .001), with longer reaction times in the hard condition.

Random effects indicated substantial between-subject variability in both
baseline RT (SD = 123ms) and condition effects (SD = 34ms), as well as
between-item variability in baseline difficulty (SD = 67ms). The correlation
between subject intercepts and slopes was -0.23, suggesting participants
with slower baseline RTs showed smaller condition effects."

R CODE FOR REPORTING:
library(lme4)
model <- lmer(RT ~ Condition + (Condition|Subject) + (1|Item), data=data)

# Fixed effects
summary(model)

# Confidence intervals
confint(model, method="Wald")

# Random effects
VarCorr(model)

# Model comparison (if relevant)
anova(model_reduced, model_full)

P-VALUES IN LMER:
lmer() doesn't report p-values by default (denominator df unclear)

OPTIONS:
1. Use lmerTest package (Satterthwaite approximation):
library(lmerTest)
model <- lmer(...)
summary(model)  # Now includes p-values

2. Use likelihood ratio tests:
drop1(model, test="Chisq")

3. Use parametric bootstrap:
library(pbkrtest)
PBmodcomp(model_full, model_reduced)

4. Report t-values with |t| > 2 as "significant" (rough rule for large df)

--------------------------------------------------------------------------------
```

9.9 POWER AND SIMULATION IN MIXED MODELS

POWER ANALYSIS:
Harder than for simple designs due to multiple variance components

FACTORS AFFECTING POWER:
- Number of subjects
- Number of observations per subject
- Between-subject variance (σ^2_{u0})
- Within-subject variance (σ^2_{ε})
- Effect size (β_1)
- Random slope variance (σ^2_{u1})

SIMULATION APPROACH:
1. Specify population parameters
2. Generate many datasets
3. Fit model to each
4. Calculate proportion of significant results = Power

R CODE (Basic Simulation):

```
library(simr)

# Start with fitted model
model <- lmer(Y ~ X + (X|Subject), data=data)

# Power analysis for effect of X
powerSim(model, fixed("X"), nsim=1000)

# Extend to larger sample size
model_extended <- extend(model, along="Subject", n=100)
powerSim(model_extended, fixed("X"), nsim=1000)

# Power curve
pc <- powerCurve(model, fixed("X"), along="Subject", nsim=1000)
plot(pc)
```

SAMPLE SIZE CALCULATION:
Iterate simulations with different n until desired power reached

REPORTING:
"Power analysis via simulation (1000 iterations) indicated that 80 subjects would provide 80% power to detect an effect of $\beta = 0.5$ with the observed variance components."

==========

10. GENERALIZED LINEAR MIXED MODELS (GLMER)

==========

10.1 WHEN TO USE GLMER

USE WHEN:
- Non-normal dependent variable
- Mixed effects needed (repeated measures, clustering)
- Binary, count, or skewed continuous data

EXAMPLES:
- Binary: Correct/incorrect responses across trials
- Count: Number of errors per session
- Skewed: Reaction times (always positive, right-skewed)

GLM vs GLMER:
- GLM: Generalized Linear Model (no random effects)
- GLMER: Generalized Linear Mixed-Effects Model (with random effects)

10.2 BINARY LOGISTIC REGRESSION

USE WHEN:
- Binary outcome (0/1, Yes/No, Success/Failure)
- Predicting probability of outcome
- Repeated measures or clustering

EXAMPLES:
- Correct vs incorrect across trials
- Choosing option A vs B
- Clinical: Improved vs not improved
- Pass vs fail over multiple tests

MODEL:
$\text{logit}(p) = \log(p/(1-p)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots$

Where:
- $p = P(Y=1)$ = probability of "success"
- logit = log-odds
- β_0 = intercept (log-odds when all X=0)
- β_1 = change in log-odds per unit change in X_1

LINK FUNCTION: logit
DISTRIBUTION: Binomial

INVERSE LOGIT (Convert log-odds to probability):
$p = \exp(\text{logit}) / (1 + \exp(\text{logit}))$

```
  = 1 / (1 + exp(-logit))

R CODE (without random effects):
model <- glm(Y ~ X1 + X2, data=data, family=binomial)

R CODE (with random effects):
library(lme4)
model <- glmer(Y ~ X1 + X2 + (1|Subject), data=data, family=binomial)

INTERPRETING COEFFICIENTS:

On log-odds scale:
β₁ = change in log-odds per unit change in X₁

On odds scale:
OR = exp(β₁) = odds ratio
• OR = 1: No effect
• OR > 1: Increases odds
• OR < 1: Decreases odds

Example:
β₁ = 0.5 → OR = exp(0.5) = 1.65
"Each unit increase in X₁ multiplies odds by 1.65 (65% increase)"

On probability scale:
Effect depends on starting probability (non-linear)
• Small effect when p near 0 or 1
• Largest effect when p near 0.5

PREDICTED PROBABILITIES:

Step 1: Calculate linear predictor
logit = β₀ + β₁X₁ + β₂X₂

Step 2: Convert to probability
p = 1 / (1 + exp(-logit))

R CODE:
predict(model, type="link")      # Log-odds scale
predict(model, type="response")  # Probability scale

ASSUMPTIONS:
✓ Binary outcome
✓ Independence (or accounted for with random effects)
✓ Linearity on log-odds scale
✓ No perfect separation
✓ Sufficient sample size (5-10 events per predictor)
```

CHECKING FIT:
• Residual deviance
• Hosmer-Lemeshow test
• ROC curve, AUC
• Classification accuracy

REPORTING:
"Logistic mixed-effects model with random intercepts for subjects. The effect of X was significant ($\beta = 0.52$, SE = 0.12, $z = 4.33$, $p < .001$, OR = 1.68). Each unit increase in X increased the odds of success by 68%."

10.3 POISSON REGRESSION (COUNT DATA)

USE WHEN:
• Count outcome (non-negative integers: 0, 1, 2, 3, ...)
• Predicting number of events
• Repeated measures or clustering

EXAMPLES:
• Number of errors per trial
• Number of words recalled
• Number of symptoms reported
• Event counts over time

MODEL:
$\log(\lambda) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots$

Where:
• λ = expected count (mean of Poisson distribution)
• $\log(\lambda)$ = log-link function
• β_1 = change in log-count per unit change in X_1

LINK FUNCTION: log
DISTRIBUTION: Poisson (assumes mean = variance)

INVERSE LOG (Convert to count):
$\lambda = \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2)$

R CODE (without random effects):
```
model <- glm(Y ~ X1 + X2, data=data, family=poisson)
```

R CODE (with random effects):
```
model <- glmer(Y ~ X1 + X2 + (1|Subject), data=data, family=poisson)
```

INTERPRETING COEFFICIENTS:

On log scale:
β_1 = change in log-count per unit change in X_1

On count scale:
Rate Ratio (RR) = $\exp(\beta_1)$
- RR = 1: No effect
- RR > 1: Increases count
- RR < 1: Decreases count

Example:
$\beta_1 = 0.3 \rightarrow RR = \exp(0.3) = 1.35$
"Each unit increase in X_1 increases expected count by 35%"

PREDICTED COUNTS:
$\hat{\lambda} = \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2)$

R CODE:
```
predict(model, type="link")      # Log scale
predict(model, type="response")  # Count scale
```

OVERDISPERSION:
Problem: Variance > Mean (violates Poisson assumption)

Detection:
Dispersion parameter = Residual Deviance / df
- If > 1.5: Overdispersion present

R CODE:
```
deviance(model) / df.residual(model)
```

SOLUTIONS FOR OVERDISPERSION:

1. Negative Binomial Regression:
Adds dispersion parameter

R CODE:
```
library(lme4)
model <- glmer.nb(Y ~ X1 + X2 + (1|Subject), data=data)
```

2. Quasi-Poisson:
Adjusts standard errors for overdispersion

R CODE:
```
model <- glm(Y ~ X1 + X2, data=data, family=quasipoisson)
```

3. Observation-level random effects:
Add random effect for each observation

R CODE:

```
data$obs_id <- 1:nrow(data)
model <- glmer(Y ~ X1 + X2 + (1|Subject) + (1|obs_id),
               data=data, family=poisson)
```

ASSUMPTIONS:
✓ Count outcome (non-negative integers)
✓ Independence (or accounted for with random effects)
✓ Mean = Variance (check for overdispersion!)
✓ Linearity on log scale

REPORTING:
"Poisson mixed-effects model with random intercepts for subjects. Dispersion parameter = 1.12, indicating adequate fit. The effect of X was significant (β = 0.32, SE = 0.08, z = 4.00, p < .001, RR = 1.38). Each unit increase in X increased expected error count by 38%."

10.4 GAMMA AND LOGNORMAL FOR SKEWED DATA

USE FOR:
- Continuous positive data (cannot be zero or negative)
- Right-skewed distributions
- Especially: Reaction times, response latencies, durations

EXAMPLES:
- Reaction time (always > 0, right skew)
- Income (positive, skewed)
- Survival time
- Waiting time

WHY NOT NORMAL LINEAR REGRESSION:
- Can predict negative values (impossible for RT)
- Residuals not normal (skewed)
- Heteroscedasticity (variance increases with mean)

TWO APPROACHES:

1. GAMMA REGRESSION (GLM family)

Model:
$\log(\mu) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

Where: μ = expected value

LINK FUNCTION: log (canonical), or inverse, or identity
DISTRIBUTION: Gamma (flexible shape for positive data)

R CODE:

```
model <- glmer(RT ~ X1 + X2 + (1|Subject),
               data=data,
               family=Gamma(link="log"))
```

Interpretation:
$\exp(\beta_1)$ = multiplicative effect on RT

Example:
$\beta_1 = 0.15 \rightarrow \exp(0.15) = 1.16$
"Each unit increase in X_1 multiplies RT by 1.16 (16% increase)"

Advantages:
- Handles skewness naturally
- Mean-variance relationship appropriate for RT
- Coefficients interpret as multiplicative effects

Disadvantages:
- Cannot handle zero values (must be strictly positive)
- Less familiar to many researchers

---

2. LOGNORMAL (LOG-TRANSFORM + LINEAR MODEL)

Method:
Transform Y to log(Y), then use linear model

Model:
$\log(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

Where: $\varepsilon \sim N(0, \sigma^2)$

R CODE:

```
model <- lmer(log(RT) ~ X1 + X2 + (1|Subject), data=data)
```

Back-transformation:
Predicted $Y = \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2)$

Interpretation:
$\exp(\beta_1)$ = multiplicative effect on RT (median)

Example:
$\beta_1 = 0.15 \rightarrow \exp(0.15) = 1.16$
"Each unit increase in X_1 multiplies median RT by 1.16 (16% increase)"

```

Advantages:
• Simpler (regular linear mixed model)
• Familiar interpretation
• Can use standard LMER tools
• Normalizes right-skewed data

Disadvantages:
• Cannot handle zero values (log(0) undefined)
• Predicts median, not mean
• Retransformation bias (predicted mean ≠ exp(predicted log))

NOTE ON ZEROS:
If data include zeros:
• Add small constant: log(Y + 0.001)
• Use different model (Poisson, Negative Binomial)

GAMMA vs LOGNORMAL:
Both appropriate for positive skewed data
• Gamma: Predicts mean, GLM framework
• Lognormal: Predicts median, simpler
• Results often similar in practice
• Try both, compare fit

ASSUMPTIONS TO CHECK:
• Positive values only
• Residuals approximately normal (on log scale)
• Homoscedasticity (on log scale)
• No influential outliers

REPORTING (LOGNORMAL):
"Reaction times were log-transformed to address right skew. A linear
mixed-effects model with random intercepts for subjects showed a significant
effect of X (β = 0.15, SE = 0.03, t = 5.00, p < .001). Each unit increase
in X was associated with a 16% increase in median RT (exp(0.15) = 1.16)."

--------------------------------------------------------------------------------

SUMMARY: CHOOSING THE RIGHT GLMER
--------------------------------------------------------------------------------
Outcome Type          Distribution    Link      Model
--------------------------------------------------------------------------------
Binary (0/1)         Binomial        logit     glmer(family=binomial)
Count (0,1,2,...)    Poisson         log       glmer(family=poisson)
Count (overdispersed) Negative Binomial log    glmer.nb()
Positive continuous  Gamma           log       glmer(family=Gamma)
Positive skewed      Lognormal       -         lmer(log(Y) ~ ...)
Proportions (0-1)    Beta            logit     glmmTMB(family=beta)
```

==========
11. BAYESIAN STATISTICS
==========

11.1 FREQUENTIST VS BAYESIAN PARADIGMS

FREQUENTIST APPROACH:
Philosophy: Probability = long-run frequency

Key concepts:
• Parameters are FIXED (unknown constants)
• Data are RANDOM (sample from population)
• P-value: P(data | H_0 is true)
• Confidence interval: Long-run coverage property
• Cannot make probability statements about parameters

Example interpretation:
"If H_0 were true and we repeated this study infinitely, we'd observe data
this extreme 5% of the time."

CANNOT say: "There's a 95% probability the parameter is in this interval"

BAYESIAN APPROACH:
Philosophy: Probability = degree of belief

Key concepts:
• Parameters are RANDOM (have probability distributions)
• Data are FIXED (what we observed)
• Updates beliefs: Prior → Likelihood → Posterior
• Posterior probability: P(parameter | data)
• CAN make probability statements about parameters

Example interpretation:
"Given the data, there's a 95% probability the parameter lies in this interval."

CAN say: "The parameter is 0.95 probable to be in [a, b]"

COMPARISON:

Aspect Frequentist Bayesian

```
Parameter         Fixed unknown            Random variable
Inference         P(data|parameter)        P(parameter|data)
Prior information Cannot incorporate       Explicitly incorporated
Probability       Long-run frequency       Degree of belief
Interval          Confidence interval      Credible interval
Multiple comparisons Need correction       No correction needed
Small samples     Can be problematic       Works well
Computation       Often simpler            Often complex (MCMC)
Interpretation    Indirect                 Direct
--------------------------------------------------------------------------------

WHEN TO USE BAYESIAN:
• Want to incorporate prior knowledge
• Small sample size
• Complex hierarchical models
• Want direct probability statements about parameters
• Sequential testing (updating as data arrive)
• Decision-making under uncertainty

--------------------------------------------------------------------------------

11.2 FUNDAMENTAL PRINCIPLES: PRIORS, LIKELIHOODS, POSTERIORS
--------------------------------------------------------------------------------

BAYES' RULE:
P(θ|Data) = [P(Data|θ) × P(θ)] / P(Data)

Or more intuitively:
Posterior ∝ Likelihood × Prior

Components:

1. PRIOR P(θ):
• Your beliefs about parameter BEFORE seeing data
• Based on: previous research, theory, expert opinion, or ignorance
• Can range from very informative to completely non-informative

2. LIKELIHOOD P(Data|θ):
• Probability of observing data given parameter value
• Same as in frequentist analysis
• Comes from your model (normal, binomial, etc.)

3. POSTERIOR P(θ|Data):
• Your updated beliefs about parameter AFTER seeing data
• Combines prior and likelihood
• What you want for inference

4. MARGINAL LIKELIHOOD P(Data):
```

• Normalizing constant (makes posterior integrate to 1)
• Usually computed via integration or MCMC

EXAMPLE: Coin Flipping

Question: What's the probability (θ) the coin is heads?

PRIOR: P(θ)
Belief before flipping: Could be fair coin
→ Beta(5, 5) distribution centered at 0.5

DATA: Flip 10 times, observe 7 heads

LIKELIHOOD: P(7 heads | θ)
Binomial: $P(Data|\theta) = C(10,7) \times \theta^7 \times (1-\theta)^3$

POSTERIOR: P(θ | 7 heads)
Combines prior and likelihood
→ Beta(5+7, 5+3) = Beta(12, 8)
→ Mean = 12/(12+8) = 0.60

Interpretation:
"Given 7/10 heads and mildly skeptical prior about bias,
we now believe θ ≈ 0.60 (60% heads)"

R CODE:
```
library(bayestestR)
prior <- distribution_beta(100, 5, 5)
likelihood_contribution <- dbinom(7, 10, prob=seq(0,1,0.01))
posterior <- distribution_beta(100, 12, 8)

plot(prior, main="Prior vs Posterior")
lines(posterior, col="red")
```

11.3 TYPES OF PRIORS
---

1. NON-INFORMATIVE (FLAT) PRIORS:
• Express complete ignorance
• All parameter values equally likely
• Let data dominate

Examples:
• Uniform[0, 1] for probability

```
• Uniform[-∞, ∞] for unbounded parameter (improper)

When to use:
• No prior knowledge
• Want results similar to frequentist
• Objective Bayesian analysis

R CODE:
prior = uniform(0, 1)  # Beta(1,1)

--------------------------------------------------------------------------------

2. WEAKLY INFORMATIVE PRIORS:
• Constrain to plausible range
• Don't overwhelm data
• Regularize estimates (prevent extreme values)

Examples:
• Normal(0, 10) for standardized effect
• Beta(2, 2) for proportion (slightly peaked at 0.5)
• Half-Cauchy(0, 2.5) for standard deviation

When to use:
• Default choice for most analyses
• Some vague domain knowledge
• Want slight regularization

R CODE:
prior = normal(0, 10)

--------------------------------------------------------------------------------

3. INFORMATIVE PRIORS:
• Based on strong prior knowledge
• Previous studies, expert opinion
• Can dominate data if very strong

Examples:
• Normal(100, 15) for IQ (strong prior)
• Beta(80, 20) for success rate (80% expected)

When to use:
• Strong prior information available
• Building on previous research
• Sequential analyses

R CODE:
prior = normal(100, 15)
```

PRIOR SENSITIVITY ANALYSIS:
Always check if results depend on prior choice!

Steps:
1. Fit model with different priors
2. Compare posteriors
3. If posteriors similar → robust to prior
4. If posteriors differ → results driven by prior (report this!)

R CODE:
```
# Fit with different priors
model1 <- stan_glm(Y ~ X, data=data, prior=normal(0,2.5))
model2 <- stan_glm(Y ~ X, data=data, prior=normal(0,10))

# Compare
posterior_interval(model1)
posterior_interval(model2)
```

11.4 BAYES FACTORS

PURPOSE: Compare evidence for two hypotheses

BAYES FACTOR (BF_{10}):
$BF_{10} = P(Data \mid H_1) / P(Data \mid H_0)$

Interpretation:
- $BF_{10} = 3$: Data are 3× more likely under H_1 than H_0
- $BF_{10} = 1/3$: Data are 3× more likely under H_0 than H_1
- $BF_{10} = 1$: Data equally likely under both hypotheses

INVERSE: $BF_{01} = 1/BF_{10}$

INTERPRETING BAYES FACTORS (Jeffreys, 1961):

BF_{10}	Evidence for H_1
> 100	Extreme evidence
30 - 100	Very strong evidence
10 - 30	Strong evidence
3 - 10	Moderate evidence
1 - 3	Anecdotal evidence
1	No evidence

```
1/3 - 1       Anecdotal evidence for H₀
1/10 - 1/3    Moderate evidence for H₀
1/30 - 1/10   Strong evidence for H₀
1/100 - 1/30  Very strong evidence for H₀
< 1/100       Extreme evidence for H₀
--------------------------------------------------------------------------------

ADVANTAGES OVER P-VALUES:
• Can support null hypothesis
• Quantifies evidence (not just "significant" or not)
• No need for correction in multiple comparisons
• Can stop data collection when evidence sufficient

EXAMPLE:
BF₁₀ = 5.2
"Data are 5.2 times more likely under H₁ than H₀ (moderate evidence for H₁)"

BF₁₀ = 0.3 (or BF₀₁ = 3.33)
"Data are 3.33 times more likely under H₀ than H₁ (moderate evidence for H₀)"

BF₁₀ = 1.2
"Data provide virtually no evidence either way"

R CODE:
library(BayesFactor)

# t-test
ttestBF(x, y)

# ANOVA
anovaBF(DV ~ Factor, data=data)

# Regression
regressionBF(Y ~ X1 + X2, data=data)

# Correlation
correlationBF(x, y)

RELATIONSHIP TO POSTERIOR ODDS:
Posterior Odds = Prior Odds × Bayes Factor

If prior odds = 1 (50/50 before), then:
Posterior odds = BF₁₀

Example: BF₁₀ = 10
Posterior odds = 10:1 in favor of H₁
Posterior probability = 10/(10+1) = 0.91 (91% probability H₁ is true)
```

```
--------------------------------------------------------------------------------

11.5 CREDIBLE INTERVALS
--------------------------------------------------------------------------------

DEFINITION:
Interval containing parameter with specified probability

95% CREDIBLE INTERVAL:
"There is a 95% probability that the parameter lies in this interval"

CONTRAST WITH CONFIDENCE INTERVAL:
Confidence: "If we repeat study 100 times, 95 intervals will contain parameter"
Credible: "Parameter is 95% probable to be in this interval"

TYPES:

1. EQUAL-TAILED INTERVAL (ETI):
• 2.5% in each tail
• Symmetric around median
• Most common

2. HIGHEST DENSITY INTERVAL (HDI):
• Narrowest interval containing 95% of posterior
• All points inside have higher density than points outside
• Preferred for skewed posteriors

R CODE:
library(bayestestR)

# ETI
ci(posterior, method="ETI", ci=0.95)

# HDI
ci(posterior, method="HDI", ci=0.95)

INTERPRETATION:
ETI: [0.3, 0.8]
"We are 95% certain the parameter is between 0.3 and 0.8"

If interval excludes 0:
"We can be 95% certain there is a positive effect"

If interval includes 0:
"We cannot be certain of the direction of the effect"

REGION OF PRACTICAL EQUIVALENCE (ROPE):
Define smallest effect size of interest (e.g., [-0.1, 0.1])
```

Check:
• % of posterior in ROPE
• If 95% CI entirely within ROPE → effectively zero
• If 95% CI entirely outside ROPE → effect present

R CODE:
```
rope(posterior, range=c(-0.1, 0.1), ci=0.95)
```

11.6 BAYESIAN WORKFLOW: PRIOR AND POSTERIOR PREDICTIVE CHECKS

BAYESIAN WORKFLOW:
1. Specify model and priors
2. Prior predictive check
3. Fit model (obtain posterior)
4. Convergence diagnostics
5. Posterior predictive check
6. Report results

PRIOR PREDICTIVE CHECK:
Simulate data from prior (before seeing real data)

Purpose:
• Check if prior is reasonable
• Ensure prior allows realistic values
• Catch mistakes in model specification

Method:
1. Sample parameters from prior
2. Generate fake data using those parameters
3. Check if fake data looks plausible

R CODE:
```
library(brms)
model <- brm(Y ~ X, data=data, prior=my_priors, sample_prior="only")
pp_check(model, type="dens_overlay")
```

Questions to ask:
• Do simulated data look anything like real data?
• Are extreme values plausible?
• Does range match reality?

If not:
→ Adjust priors

```
--------------------------------------------------------------------------------

POSTERIOR PREDICTIVE CHECK (pp_check):
Simulate data from posterior (after seeing real data)

Purpose:
• Check model fit
• Identify model misspecification
• Ensure model can generate data like observed

Method:
1. Sample parameters from posterior
2. Generate fake data (y_rep) using those parameters
3. Compare y_rep to observed data (y)

R CODE:
library(brms)
pp_check(model, type="dens_overlay", ndraws=100)

TYPES OF pp_checks:

1. Density overlay:
Overlay simulated data densities on observed density

pp_check(model, type="dens_overlay")

Good fit: y_rep densities overlap with y
Bad fit: y_rep systematically different from y

2. Test statistic:
Compute statistic (mean, SD, max, etc.) for observed and replicated data

pp_check(model, type="stat", stat="mean")

Good fit: Observed statistic within range of replicated
Bad fit: Observed statistic in tail of replicated distribution

3. Interval predictions:
Plot data with prediction intervals

pp_check(model, type="intervals")

4. Error scatter:
Plot residuals

pp_check(model, type="error_scatter")

INTERPRETATION:
```

If pp_check shows poor fit:
• Model missing important predictors
• Wrong distribution family
• Need non-linear terms
• Outliers present

Actions:
• Add predictors
• Transform variables
• Use different likelihood
• Use mixture model

11.7 MCMC SAMPLING AND DIAGNOSTICS

MCMC (Markov Chain Monte Carlo):
Algorithm for sampling from posterior when can't compute analytically

COMMON MCMC ALGORITHMS:
• Metropolis-Hastings
• Gibbs sampling
• Hamiltonian Monte Carlo (HMC) - used by Stan
• No-U-Turn Sampler (NUTS) - adaptive HMC

HOW MCMC WORKS:
1. Start at random parameter value
2. Propose new value
3. Accept/reject based on posterior probability
4. Repeat thousands of times
5. Sequence of samples approximates posterior

TERMINOLOGY:
• Chain: Sequence of samples
• Iteration: Each step in chain
• Warmup/Burn-in: Initial samples discarded (before convergence)
• Thinning: Keep every k-th sample (reduce autocorrelation)

RUNNING MCMC:
Typical settings:
• 4 chains (independent runs)
• 2000 iterations per chain
• 1000 warmup iterations
• Total: 4000 posterior samples (4 chains × 1000 post-warmup)

R CODE (brms):
library(brms)

```
model <- brm(Y ~ X, data=data,
             chains=4, iter=2000, warmup=1000, thin=1)

--------------------------------------------------------------------------------

CONVERGENCE DIAGNOSTICS:

1. TRACE PLOTS:
Plot parameter value vs iteration

Good:
• Fuzzy caterpillar (random fluctuation around mean)
• All chains overlap
• No trends or patterns

Bad:
• Chains don't overlap
• Trends (still exploring space)
• Stuck at one value

R CODE:
plot(model, type="trace")

2. R-HAT (Gelman-Rubin statistic):
Compares within-chain and between-chain variance

Formula:
R̂ = √(Var_total / Var_within)

Interpretation:
• R̂ = 1.00: Perfect convergence
• R̂ < 1.05: Acceptable (some say < 1.01)
• R̂ > 1.05: Chains haven't converged

Rule: R̂ should be < 1.05 for ALL parameters

If R̂ > 1.05:
• Run longer (more iterations)
• Check for coding errors
• Reparameterize model
• Use better priors

R CODE:
summary(model)  # Check Rhat column

3. EFFECTIVE SAMPLE SIZE (ESS):
Number of "independent" samples (after accounting for autocorrelation)

```

Two versions:
• ESS_bulk: For mean, median
• ESS_tail: For quantiles, CI bounds

Rule: ESS should be > 400 (some say > 1000)

Why it matters:
• High autocorrelation → fewer independent samples
• Low ESS → less precise posterior estimates

If ESS < 400:
• Run longer
• Thin more aggressively
• Reparameterize

R CODE:
```
summary(model)  # Check Bulk_ESS and Tail_ESS columns
```

4. AUTOCORRELATION:
Correlation between successive samples

Plot: Autocorrelation vs lag

Good:
• Drops to near zero quickly (<10 lags)

Bad:
• Stays high for many lags (slow mixing)

R CODE:
```
plot(model, type="acf")
```

PRACTICAL CHECKLIST:
✓ All $\hat{R}$ < 1.05
✓ All ESS > 400 (preferably > 1000)
✓ Trace plots look like fuzzy caterpillars
✓ Autocorrelation drops quickly
✓ pp_check looks reasonable

If any fail:
• Run more iterations
• Check priors (too vague?)
• Reparameterize model
• Center/scale predictors

11.8 BAYESIAN VERSIONS OF COMMON TESTS

BAYESIAN T-TEST:

Frequentist: t.test(x, y)
Bayesian: ttestBF(x, y)

Output:
- Bayes Factor (BF_{10})
- Posterior distribution of effect size (δ)
- Credible interval for effect

R CODE:
```
library(BayesFactor)
bf <- ttestBF(x, y)
bf  # Print BF
posterior <- posterior(bf, iterations=10000)
summary(posterior)
```

Interpretation:
"BF_{10} = 8.3 indicates moderate evidence for a difference.
The posterior mean effect size is δ = 0.52, 95% CI [0.18, 0.87]."

BAYESIAN ANOVA:

Frequentist: aov(DV ~ Factor, data=data)
Bayesian: anovaBF(DV ~ Factor, data=data)

Output:
- BF for each model vs null
- BF for comparisons between models

R CODE:
```
bf <- anovaBF(DV ~ FactorA * FactorB, data=data)
bf  # Compare all models

# Specific comparison
bf[4] / bf[3]  # Main effects vs Main + Interaction
```

Interpretation:
"BF for main effects model vs null = 25.4 (strong evidence).
Adding interaction: BF = 2.1 (anecdotal evidence for interaction)."

BAYESIAN REGRESSION:

Frequentist: lm(Y ~ X1 + X2, data=data)
Bayesian: stan_glm(Y ~ X1 + X2, data=data)

Output:
• Posterior distributions for all coefficients
• Credible intervals
• Model fit (R^2, Bayes R^2)

R CODE:
```
library(rstanarm)
model <- stan_glm(Y ~ X1 + X2, data=data,
                  prior=normal(0, 2.5))
summary(model)
posterior_interval(model, prob=0.95)
```

Interpretation:
"Posterior mean for X1 is β = 2.3, 95% CI [1.1, 3.5].
We are 95% certain the effect of X1 is positive."

BAYESIAN MIXED MODELS:

Frequentist: lmer(Y ~ X + (X|Subject), data=data)
Bayesian: brm(Y ~ X + (X|Subject), data=data)

Output:
• Posterior distributions for fixed and random effects
• Variance components
• Credible intervals

R CODE:
```
library(brms)
model <- brm(Y ~ X + (X|Subject), data=data,
             prior=c(prior(normal(0,5), class=b),
                     prior(cauchy(0,2), class=sd)))

summary(model)
pp_check(model)
```

Interpretation:
"Fixed effect of X: β = 3.2, 95% CI [2.1, 4.3].
Between-subject SD for slopes: 1.8, 95% CI [1.2, 2.6]."

BAYESIAN CONTINGENCY TABLES (INDEPENDENCE):

Frequentist: chisq.test(table(x, y))
Bayesian: contingencyTableBF(table(x, y), sampleType="indepMulti")

Output:
• BF_{10} for association vs independence

R CODE:
```
library(BayesFactor)
tab <- table(data$gender, data$choice)
bf <- contingencyTableBF(tab, sampleType="indepMulti")
bf
```

Interpretation:
"BF_{10} = 12.5 indicates strong evidence that gender and choice are associated."

ADVANTAGES OF BAYESIAN VERSIONS:
• Direct probability statements about parameters
• Can support null hypothesis
• No p-value threshold
• Incorporate prior knowledge
• Uncertainty quantification
• More intuitive interpretation

DISADVANTAGES:
• Computationally intensive
• Requires understanding of priors
• Less familiar to many researchers
• Longer run times
• Requires more expertise to implement properly

==========
END OF COMPREHENSIVE SUMMARY
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This document provides:
✓ All statistical tests organized by topic
✓ When to use each test (scenarios and examples)
✓ Assumptions for each test
✓ Formulas and calculations
✓ R code for implementation
✓ Interpretation guidelines
✓ Decision trees for test selection
✓ Comparison tables

Use this as a reference guide for:

- Choosing appropriate statistical tests
- Checking assumptions
- Interpreting results
- Exam preparation

Remember:

1. Always check assumptions before using a test
2. Consider effect sizes, not just p-values
3. Visualize your data first
4. Report confidence/credible intervals
5. Consider power and sample size
6. Document all decisions and model specifications

Good luck with your studies and analyses!